

CANDOR: SCORE FOLLOWING AS SEQUENTIAL SELECTION OVER NOTE HYPOTHESES

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ABSTRACT

We formulate real-time score following on sheet-music images as sequential selection: at each instant the page affords a set of note hypotheses, and following the performance means choosing among them under a temporal constraint, rather than regressing a coordinate. CANDOR realizes this with three parts, a candidate generator, a parameter-free transition prior over displacements, and a 23k-parameter permutation-invariant scorer trained listwise on synthesized audio. On real room-microphone piano recordings CANDOR tracks 94.4% of onsets within 0.5 s and runs in 1.25 ms per step on one CPU thread. The formulation explains where accuracy on real audio is lost. A candidate oracle shows that a correct hypothesis is always present and a causal selector could reach 99.6%, so the synthetic-to-real gap is a decision problem, not a perception problem; consistent with this, three different image representations for the scorer, including a self-supervised vision transformer and none at all, give identical accuracy, whereas removing the temporal history costs 7.2 points. Because selection is separable from proposal, one trained selector improves every candidate generator we attach it to, by 20.3, 11.0 and 7.9 points. We further report that accuracy on synthesized validation audio is nearly uninformative about accuracy on real audio ($r = 0.16$ over 48 checkpoints), and that the forward-motion prior which makes selection work is also what makes score discontinuities hard.

Index Terms— Score following, sheet music, real-time alignment, set prediction, music information retrieval

1. INTRODUCTION

Score following, the real-time alignment of a live performance to its score, underlies automatic page turning [1] and computer accompaniment [2]. Classical systems align audio to a *symbolic* score and infer the position over a discrete set of score states with hidden (semi-)Markov models [3, 4] or online time warping [5]. Most music, however, exists only as printed or scanned sheets, and optical music recognition is not reliable enough to produce that symbolic score automatically [6]. A decade of work therefore follows the performance directly on the sheet *image* [7, 8, 9, 10, 11, 12]: the input is a page of pixels and a stream of audio, the output is where on that page

the musician is.

How the field does this today. Three families have emerged, and how they differ matters for what follows. *Dense localizers* treat the page as a canvas and paint on it: MM-Loc [7] and CUNet [9] emit an audio-conditioned heatmap over the whole image and read a coordinate off its centre of mass. *Agents* treat it as control: a reinforcement-learning policy [8] moves a reading head and is rewarded for staying aligned. *Audio-conditioned detectors* treat it as a query: CYOLO [10] and its multi-resolution successor CYOLO-SB [11] encode the recent audio into a vector, modulate a visual backbone with it through FiLM [13], and run object detection, so that the audio in effect asks *which part of this page do I sound like*. All three commit to the same thing, that the answer is a coordinate the network must produce, all three train on audio synthesized from the score, and all three lose much of their accuracy on real recordings, the field’s main open problem: CUNet falls from 85.5% of onsets tracked within 0.5 s to 22.4%, and CYOLO-SB, the most robust, from 89.3% to 79.9% [12].

Our starting point. We argue the natural formulation is not regression but *selection*. A page of music affords a finite, discrete set of places the performance could be at, one per notehead, and following the score is choosing among them, again at every instant, subject to the choices forming a plausible trajectory. This is what symbolic followers have always done over score states; on a sheet image the obstacle has been that the state set is not given, it must come from the pixels. Our observation is that a detector already produces it. CYOLO-SB emits thousands of note boxes per instant and discards all but the most confident, and that discarded set is exactly the hypothesis space a selection formulation needs.

One measurement shows what is at stake. Keeping 256 hypotheses per instant, we find that on real room-microphone piano recordings a hypothesis within the 0.5 s tolerance is present at every one of 4,149 scored onsets, and that an oracle which must commit one step at a time and may never move backwards tracks 99.6% of them. Reporting the most confident hypothesis, which is what the detector does, tracks 80.0% (Fig. 1). Nineteen of the twenty points lost to real audio are lost in the decision, not in the perception. Under a regression view this is invisible, because there is no set over which to take an oracle.

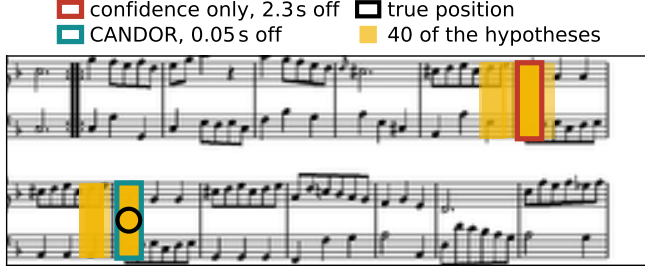


Fig. 1. One instant of a real room-microphone recording. Yellow: 40 of the note hypotheses the page affords at this instant. Reporting the most confident (red) lands on the wrong staff; CANDOR (teal) selects a lower-confidence hypothesis on the true note (ring). The right answer was in the set and was not chosen.

CANDOR is a score follower built on this formulation: a candidate generator, a transition prior with no trained parameters, and a small permutation-invariant scorer that judges each hypothesis against the others in its instant. Because selection is separable from proposal, the generator is an interchangeable component; we attach one trained selector to three published detectors and it improves each, which is a property of the formulation rather than a tuning result.

The concurrent CODA [12] also abandons regression, and the contrast sharpens ours. It selects over *given* structure, cascading system then bar then note over the layout regions of an annotated score, with a network trained end to end. We select over *proposed* structure, the note hypotheses a frozen detector already emits, so no layout annotation is read at test time and no perception model is retrained.

Our contributions are: (i) a formulation of sheet-image score following as sequential selection over note hypotheses, with a candidate-oracle analysis locating the synthetic-to-real gap in the decision; (ii) CANDOR, a tracker realizing it, reaching 94.4% of onsets within 0.5 s on real recordings at 1.25 ms per instant and transferring unchanged across candidate generators; (iii) three findings the formulation exposes and regression hides: the image representation the selector reads is irrelevant while its temporal history is essential, synthetic validation accuracy barely ranks models for real audio, and the forward-motion prior that makes selection work is what makes discontinuities hard.

2. SCORE FOLLOWING AS SEQUENTIAL SELECTION

2.1. Hypotheses

A performance is processed in instants $t = 1, \dots, T$. At each instant a candidate generator reads the audio so far and the current page and emits note hypotheses $b_i = (x_i, y_i, w_i, h_i)$ with confidence o_i ; we keep the $K = 256$ most confident, $\mathcal{C}_t$.

Staves are concatenated along one horizontal axis, the unrolled coordinate u_i , so that positions on either side of a line break are adjacent. Any note detector can serve; we use released ones [10, 11], which emit 8,112 hypotheses per instant on a 52×52 grid, and never update their weights. Reporting the most confident hypothesis,

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} o_i, \quad (1)$$

is the degenerate selector, and is what a detection-based regressor does. CANDOR reads $\mathcal{C}_t$ and its own previous choice $\hat{u}_{t-1} = u_{\hat{i}_{t-1}}$, and nothing else.

2.2. Temporal constraint

Selections must form a plausible trajectory. Between instants Δ_t frames apart the performer advances by a roughly predictable amount, so we score the displacement $d_i = u_i - \hat{u}_{t-1}$ with a floored Gaussian whose mean grows with elapsed time,

$$\log \pi_i = \max \left(-\frac{(d_i - \mu \rho_t)^2}{2\sigma^2}, J \right), \quad (2)$$

with $\rho_t = \text{clip}(\Delta_t / \Delta_0, 0.2, 8)$ and $\Delta_0 = 5$ frames, the mean inter-onset gap in training. The floor J leaves jumps possible at bounded cost. None of (μ, σ, J) is learned. Combined with confidence, $h_i = \log o_i + \log \pi_i$, this alone is a competent selector (Table 2).

2.3. Set scorer

The prior treats hypotheses independently given the history. What it cannot express is comparative: whether *this* box, among the ones on offer right now, looks like a real note-head. Each hypothesis is therefore described by $\phi_i \in \mathbb{R}^{33}$ holding confidence relative to the instant's best, box geometry, motion relative to the history, relation to any coarse regions the generator emits, and a neighbourhood group (how many hypotheses lie within 25 and 100 px, spacing to the nearest on each side, rank in reading order, size against the instant's median). We also read the 128-dimensional feature f_i from which the generator scored the box. Re-scoring detections against each other has precedent in vision [14]; a permutation-invariant network [15] embeds the set and pools a context every hypothesis sees:

$$e_i = g([\phi_i; \text{ReLU}(W f_i)]), \quad c = \left[\frac{1}{K} \sum_j e_j; \max_j e_j \right], \quad (3)$$

$$s_i = r([e_i; c]), \quad (4)$$

with g and r two-layer perceptrons (64, 32 and 64, 1 units) and $W \in \mathbb{R}^{64 \times 128}$. The maximum is per dimension, so each hypothesis is judged against its competitors. The scorer has 23,203 parameters.

2.4. Training and decoding

Selection is supervised listwise. With ε_i the error in frames of hypothesis i , the target is $q_i \propto \exp(-\varepsilon_i/\tau)$, $\tau = 3$, and the loss is

$$\mathcal{L} = - \sum_{i \in \mathcal{C}_t} q_i \log \frac{\exp s_i}{\sum_{j \in \mathcal{C}_t} \exp s_j}, \quad (5)$$

so the scorer is trained to rank a set rather than to regress a coordinate. Training runs on the MSMD training pieces [16], synthesized and convolved with measured room impulse responses [17]. History features come from the previous instant’s selections, perturbed on 30% of samples by Laplace noise of scale 30 px. We use AdamW ($3 \cdot 10^{-3}$, cosine, 40 epochs, batch 256). At each instant CANDOR selects

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} \beta s_i + (1 - \beta) h_i, \quad (6)$$

and $\hat{i}_t$ becomes the history of instant $t + 1$. Decoding is causal and greedy: one forward pass over 256 hypotheses, 1.25 ms on a single CPU thread. Training on the residual $s_i - h_i$ and selecting $\arg \max_i (s_i + h_i)$ is Eq. 6 at $\beta = 0.5$ and removes all four scalars.

3. EXPERIMENTS

Data and metric. Selectors are trained on 353 MSMD pieces [16] and validated on 19. The evaluation set is 16 pieces, 25 pages and 4,149 onsets played on a Yamaha AvantGrand N2 and captured with a room microphone [8]; a held-out set of 80 further pieces is rendered with a real impulse response and noise from 12 to 0.5 dB SNR. We report the ratio of onsets whose selected position maps to within 0.05 to 5 s of the true time, 0.5 s primary, with 95% intervals bootstrapped over pieces [18].

The four scalars (μ, σ, J, β) are fitted leave-one-piece-out: for each evaluation piece they are chosen on the other 15, pieces weighted equally, so none influences its own decoding. Every CANDOR number below is cross-validated this way and the residual variant needs no fit at all. Seeds move results by about a point, so rows are seed means with deviations.

3.1. Selection is separable from proposal

Table 1 reports CANDOR on real recordings and Table 2 decomposes it. The same formulation, with the selector retrained on whichever hypotheses the generator emits, holds across three generators of very different quality. It is worth 20.3 points on the weakest, whose training data is entirely public, lifting it past where the strongest generator begins, and reaches 94.4% on the strongest. A selector fitted to one kind of proposal does not need the proposals to be good; it needs them to contain the answer, which the oracle says they do.

Two rows keep the claim honest. The parameter-free prior alone accounts for roughly half the gain, so the learned component is not carrying the result by itself, and the residual variant,

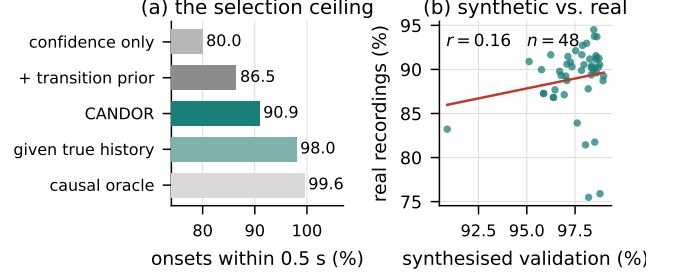


Fig. 2. (a) Every step from the degenerate selector to the ceiling on the real recordings. CANDOR to *given true history* is drift; above it is ranking. (b) Each point is one of 48 trained selectors: accuracy on synthesised validation audio against accuracy on the real recordings.

which fits no scalar to any evaluation data, reaches 88.7%. Selection is also cheap: 23k parameters and 1.25 ms per instant against 12.8 ms on a GPU for the closest comparable system [12].

3.2. What selection needs, and what it does not

Not pixels. Feeding the selector the generator’s backbone features, DINOv2 patch features [19] at the same grid, or no image features at all gives 90.9 in all three cases. On synthesized validation audio the same three differ by 3.4 points, so the features are learnable and useful there and worth nothing where it counts. Replacing the generator’s audio encoder is no better: a MERT tower [20] costs more than 40 points, and **Mamba pending**. Perception is not the binding constraint, which is what the oracle predicted.

But history. Removing the features computed against the previous selection costs 7.2 points, the largest single ablation here. Teacher forcing attributes about two thirds of the remaining error to drift: supplied the true previous position the same selector reaches 98.0% against the causal oracle’s 99.6% (Fig. 2a). Selection is sequential, and it is the sequence that is hard.

Synthetic accuracy does not rank selectors. Across 48 trained checkpoints, accuracy on synthesized validation audio correlates $r = 0.16$ with accuracy on real recordings (Fig. 2b); the best-validation checkpoint scores 89.3 on real audio, below a selector reading no image features at all. A reverberant held-out set does better ($r = 0.80$) and still mispicks. This is why the four scalars are fitted on the target recordings under cross-validation rather than on a synthetic proxy, and it is a caution for the field.

3.3. Where the temporal constraint hurts

Eq. 2 assumes forward motion, which is why selection works and also why a repeat or a jump breaks it. We splice three

Table 1. Real piano recordings, 16 pieces and 4,149 onsets captured with a room microphone. Ratio of onsets tracked within each error threshold. [†] generator trained with additional proprietary recordings, public checkpoint. *Layout*: needs system and bar annotations at test time. *Trains*: what the method fits on the target task. Published figures are as reported in [12].

	Method	Onset error threshold (s)					Layout	Trains
		≤0.05	≤0.1	≤0.5	≤1.0	≤5.0		
ours	CANDOR, proposed structure, CYOLO	.	.	.916	.	.	no	23k selector
	CANDOR, proposed structure, CYOLO-SB	.	.	.909	.	.	no	23k selector
	CANDOR, proposed structure, CYOLO-SB+A [†]	.	.	.944	.	.	no	23k selector
prior work	MM-Loc [7], dense	.364	.406	.585	.611	.735	no	1.2M end to end
	RL agent [8], control	.185	.200	.476	.603	.901	no	policy
	CUNet [9], dense	.113	.125	.224	.266	.443	no	4.5M end to end
	CYOLO [10], query	.563	.581	.712	.749	.919	no	1.5M detector
	CYOLO-SB [11], query	.610	.630	.799	.832	.960	no	1.5M detector
	CYOLO-SB+A [†] [11]	.682	.706	.865	.891	.981	no	1.5M detector
	CODA [12], given structure	.721	.743	.883	.914	.987	yes	2.0M end to end

Table 2. Components of CANDOR, all cross-validated per piece on the real recordings at 0.5 s. *argmax* is Eq. 1 over the same hypotheses, *prior* adds Eq. 2, *full* adds the scorer.

	<i>n</i>	<i>argmax</i>	<i>prior</i>	<i>full</i>
<i>Candidate generator</i>				
CYOLO [10]	3	71.3	72.5	91.6 ± 1.4
CYOLO-SB [11]	6	80.0	86.5	90.9 ± 0.6
CYOLO-SB+A [†]	3	86.5	90.8	94.4 ± 0.9
<i>Selector, on CYOLO-SB hypotheses</i>				
residual, no fitted scalars	3	80.0	86.5	88.7 ± 0.9
without temporal history	3	80.0	86.5	83.7 ± 0.4
<i>Image representation the selector reads</i>				
generator backbone (default)	6	80.0	86.5	90.9 ± 0.6
DINOv2 [19]	1	80.0	86.5	90.9
none	1	80.0	86.5	90.9
<i>Audio encoder inside the generator, retrained</i>				
MERT [20]	3	33.4	31.9	46.2 ± 9.1
Mamba [21]	3	.	.	.

discontinuities into each performance at random destinations, following the protocol of [12], re-running the generator on the spliced audio so its conditioning is genuinely wrong afterwards, and excluding jumps across a page boundary where our state resets anyway, leaving 41.

Table 3 reports a failure. CANDOR is worse after a jump than selecting by confidence alone, 0.15 against 0.22 of onsets tracked over the next five seconds, because the prior pins the trajectory to a stale position. Suppressing it on detected silence recovers part of that when a break exists and nothing when it does not; triggering instead on the selector’s own confidence needs no break and works in both cases, but only returns to the confidence baseline. A prior worth eleven points on continuous playing is strong enough to be wrong at a discontinuity, and handling both in one selector is open.

Table 3. Recovery from 41 spliced discontinuities. *rec@1*: fraction first tracked correctly within 1 s. *post*: onsets tracked within 1 s over the following 5 s.

Selector	break present		no break	
	rec@1	post	rec@1	post
confidence only (1)	0.32	0.22	0.27	0.22
CANDOR	0.12	0.15	0.10	0.15
+ silence trigger	0.27	0.19	0.10	0.15
+ confidence trigger	0.15	0.22	0.20	0.23

3.4. Robustness and budget

Under a real impulse response and noise on 80 held-out pieces the scorer beats the prior at every level from 12 to 0.5 dB, by less as the SNR falls. On 94 synthetic test pieces, where confidence alone selects .893 at 0.5 s, CANDOR reaches .955. Halving the budget to $K = 128$ costs 0.5 points and the correct hypothesis sits at rank 27 at the 90th percentile: the tail is load-bearing.

4. CONCLUSION

Posing sheet-image score following as selection over note hypotheses makes the synthetic-to-real gap measurable and mostly closable: the answer is already in the set, and a 23k-parameter selector finds it 94.4% of the time at 1.25 ms per instant, transferring across generators because selection and proposal are separate concerns. What it needs is its own history, not better pixels, and the constraint it relies on is what fails at a discontinuity. The correct page is given, all performances are piano, and 16 pieces cannot resolve a point or two. Pitch agreement is the natural next history-independent cue.

5. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical study that uses only publicly available data; no ethical approval was required.

6. ACKNOWLEDGMENTS

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